The Æther-flow Interpretation of Relativity

Flagship Public Article

Abstract

This article is a single-entry public synthesis of Æther-flow Interpretation of Relativity built on the active exact-closure manuscript sequence. Its job is not to replace that sequence, but to make it readable to first-time visitors, technically trained readers, and future collaborators. The central public message is simple. The flagship deliverable of the repository is already the overview-first exact-closure package centered on the *Exact-Closure Sequence Overview*. Within that package, the effective gravitational dynamics are adopted to be exactly those of general relativity with universal matter coupling, while the Æther / Æther-flow ontology supplies the interpretive and conceptual setting.

The article makes five points explicit. First, the active theory statement is exact relativistic closure, not a low-energy deformation of GR. Second, the Æther / Æther-flow language is retained in disciplined ontological form through the notions of underlying substrate, intrinsic ordered motion, observed three-dimensional space as local experiential slice, and S-time as experienced order of change. Third, the public claim boundary is strict: GR is adopted exactly in the active package, but a completed first-principles substrate derivation is not yet claimed. Fourth, the overview, exact-closure note, and five companion manuscripts form the ordered scientific core of the package. Fifth, the present article is a release-facing wrapper built on that core, while the larger derivational bridge archive remains available for transparency and future work but is downstream and optional relative to it; the current frozen primitive-reservoir line is now recorded only as a scoped no-go on its frozen bridge object rather than as a prompt for another same-output relay.

1. Introduction

The repository now has a stable public-facing theory package. That fact changes the right mode of presentation. The project no longer needs to be introduced primarily as ontology awaiting its first disciplined theory statement. The exact theory statement already exists in the overview-first exact-closure sequence. What is now needed is a public article that explains that package cleanly, states the claim boundary honestly, and helps readers know where to start.

This article therefore has a narrower role than the companion manuscripts. The *Exact-Closure Sequence Overview* remains the canonical front door to the package itself, and the *Exact-Closure Note* remains the shortest standalone anchor. The present article is instead a public-facing synthesis. It explains what the active package is, what it does and does not claim, how the companion papers fit together, and why the larger derivational archive should be read only afterward.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the

local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The immediate public task is therefore not another bounded bridge installment on the frozen current line. It is to package, review, and present the exact-closure sequence as the achieved flagship result. That stance depends on two fixed facts already recorded in the active sequence: the benchmark package is the positive exact-theory statement already in hand, and the current bounded primitive-reservoir line has already ended in the scoped verdict `Not Derived On Current Line` on its frozen bridge object.

2. Public Role of This Article

Definition 1 (Flagship public article)**.** *A flagship public article is a release-facing manuscript that introduces the active exact-closure package in one readable document, preserves the fixed claim boundary of the package, and directs readers into the full manuscript sequence without reopening the underlying theory line.*

Proposition 1 (Sequence-preserving role)**.** *The present article is secondary to the exact-closure sequence overview in scientific authority and sequence order, but primary as a public orientation document for readers who need a single coherent entry article.*

This distinction matters. If the public article were allowed to replace the exact-closure overview, the package could drift into a looser or more promotional presentation. If it were omitted entirely, first-time readers would be left to infer the status of the project from a distributed manuscript tree. The correct middle position is to let the package remain the source of truth while the present article functions as its public guide.

3. Ontology in Disciplined Form

The conceptual core of Æther / Æther-flow framework is ontological rather than deformation-first. The active vocabulary is:

- Æther: the underlying four-dimensional substrate of reality;
- Æther-flow: the intrinsic ordered motion of that substrate;
- observed three-dimensional space: the local experiential slice of the deeper substrate;
- S-time: the experienced order of change arising from matter, light, and the Æther-flow;
- observed expansion: the three-dimensional appearance of deeper four-dimensional ordered motion.

These terms are intended to discipline the interpretive proposal, not to overstate the dynamical derivation. The Æther-flow is not introduced as a naive vector wind in ordinary observed space. The observed world is not identified with the whole ontology. The point of the framework is instead

that the observer-accessible relativistic world may be interpreted as the experienced appearance of deeper ordered substrate structure.

Remark 1. *At the active public stage, this ontology is real as interpretive structure, but it is not yet advertised as a completed microphysical derivation of the relativistic sector.*

4. Exact Relativistic Theory Statement

Definition 2 (Æther-flow Interpretation of Relativity)**.** *Æther-flow Interpretation of Relativity is the exact-closure, GR-consistent theory of Æther / Æther-flow framework in which the Æther / Æther-flow ontology is retained while the effective gravitational dynamics are exactly those of general relativity with universal matter coupling.*

Its effective action is

$$S_{\text{eff}} = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g}\, R + S_{\text{matter}}[g, \psi], \tag{1}$$

with field equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{2}$$

the standard null condition

$$0 = g_{\mu\nu}\, dx^\mu dx^\nu, \tag{3}$$

and the standard proper-time rule

$$d\tau^2 = -\frac{1}{c^2} g_{\mu\nu}\, dx^\mu dx^\nu. \tag{4}$$

Proposition 2 (Operational identity with GR)**.** *For fixed matter sector, initial data, and boundary conditions, the effective observable content of Æther-flow Interpretation of Relativity is exactly that of general relativity.*

This proposition is the center of the exact-closure package. The project does not currently claim a verified low-energy deformation of GR. It claims that the ontology can be retained while the effective gravitational law remains exactly Einsteinian.

5. Adoption, Interpretation, and Open Derivation

Definition 3 (Adoption)**.** *Adoption is the use of an already established effective dynamics, here GR with universal matter coupling, without claiming that the dynamics have already been derived from deeper substrate variables.*

Definition 4 (Derivation)**.** *Derivation is a first-principles recovery of the effective relativistic sector from explicit substrate variables, together with whatever mode control, uniqueness, and closure arguments are needed to justify that recovery.*

The active public claim of Æther-flow Interpretation of Relativity is adoption with exact closure, not completed derivation. That boundary should be stated positively rather than defensively. It

means that the effective theory statement is already disciplined and complete at the operational level, while the deeper foundational recovery problem remains open.

The package therefore establishes the following.

1. The ontological vocabulary is fixed in disciplined form.
2. The exact effective relativistic law is fixed as Einsteinian with universal matter coupling.
3. The package carries no separate established low-energy non-GR observable sector.
4. The project distinguishes clearly between interpretation, adoption, and derivation.

The package also does *not* claim the following.

1. It does not claim that GR has already been derived from substrate microphysics.
2. It does not claim that ontology by itself settles the microphysical problem.
3. It does not claim that the larger bridge archive has already produced a completed Einstein derivation.
4. It does not claim that the present public package must be read through the bridge archive before it can count as a theory statement.

6. Related Work and Literature Positioning

A public-facing benchmark package should not leave its literature category to guesswork. The clean comparison is therefore explicit.

1. **Standard GR with interpretive language.** Relative to ordinary GR, the active benchmark package keeps the same low-energy equations and observable content. The difference is not a second field equation, a new low-energy graviton sector, or a modified matter route. It is the disciplined interpretive overlay supplied by the Æther / Æther-flow ontology and by the congruence-based geometric reading of that ontology.
2. **Einstein-aether theory.** Einstein-aether theory introduces a dynamical unit timelike vector field at observer level and therefore a genuine preferred-frame sector with additional coupled propagating modes (Jacobson and Mattingly, 2001; Jacobson and Mattingly, 2004). The active benchmark package does not do that. It claims no surviving observer-level aether vector field, no extra low-energy graviton sector, and no modified Einsteinian observer law.
3. **Thermodynamic or other emergent-gravity derivations.** Programs such as Jacobson's derivation of the Einstein equation as an equation of state (Jacobson, 1995) are closer in ambition to the downstream derivation program than to the benchmark package itself. Their category is a derivational explanation of GR, not an exact-GR interpretive package already separated from its open derivation burden.
4. **Sakharov-style induced gravity.** Sakharov-style induced-gravity proposals likewise ask how the Einstein-Hilbert sector may arise effectively from deeper microphysics (Sakharov, 1967/1991 English reprint). That is conceptually adjacent to the project's open foundational ambition, but it is not the same as claiming that such an induced derivation has already been completed here.

5. **Hořava gravity and other preferred-foliation alternatives.** Hořava gravity proposes a different dynamical framework with preferred foliation structure and altered mode content, even when it is designed to approach relativistic behavior in an infrared limit (Hořava, 2009). The active benchmark package is not of that type, because its operational content is exactly GR rather than a modified low-energy gravity law awaiting experimental separation from GR.

Proposition 3 (Most accurate current literature label)**.** *At current scope, Æther-flow Interpretation of Relativity is best described as an exact-GR interpretive package plus an open emergent-gravity research program, not as an already completed low-energy preferred-frame alternative to GR.*

This proposition sharpens the only defensible novelty statement. The project does not earn novelty merely by saying that GR might emerge from deeper structure, because that ambition already belongs to an established emergent-gravity literature. It also does not earn novelty by introducing a new low-energy preferred-frame field theory, because the active benchmark package explicitly refuses that move. If a narrower novelty claim survives review, it lies instead in the combination of disciplined ontology, exact-GR benchmark packaging, benchmark-gated derivation architecture, and willingness to preserve a scoped no-go rather than relabel the failed frozen line as progress.

Remark 2. *Representative comparison points are Einstein's formulation of GR (1915/1916), Einstein-aether theory (Jacobson and Mattingly, 2001; Jacobson and Mattingly, 2004), Jacobson's thermodynamic derivation of the Einstein equation (1995), Sakharov-style induced gravity (1967/1991 English reprint), and Hořava gravity (2009).*

7. The Flagship Exact-Closure Package

The active public package should be read in the following fixed order.

Order	Manuscript	Role
1	Exact-Closure Sequence Overview	Canonical front door fixing the package role, reading order, and claim boundary.
2	Exact-Closure Note	Short standalone anchor stating the exact-closure theory in its briefest complete form.
3	Foundations	Ontological vocabulary, framework intent, and the exact-closure relation to GR.
4	Dynamics	Effective action, matter coupling, weak-field structure, and benchmark exact law.
5	Consistency	Gauge structure, degree-of-freedom counting, and effective health conditions.
6	Relativistic Recovery	Exact relation to GR and local relation to SR.
7	Flow Geometry	Congruence-based geometric dictionary for reading Æther-flow inside exact relativistic structure.

Corollary 1 (Public packaging rule)**.** *The overview-first exact-closure sequence is the flagship deliverable of the repository and should be presented before any downstream derivational continuation.*

This order is not arbitrary. It places the finished effective theory statement before the open foundational program. Public accessibility and scientific honesty both improve when that order is held fixed. The present flagship public article is built on that ordered core as a release-facing synthesis. It is not an eighth scientific source that reorders the package, and it does not displace the overview as the canonical front door. The current-line obstruction / no-go note belongs downstream of this package in exactly that sense: it is a bounded verdict on a frozen bridge object, not a replacement front door and not a reason to blur the benchmark package back into open derivational work.

8. Relation to the Downstream Derivational Bridge Archive

The repository also contains a large derivational bridge archive. That archive remains important. It documents genuine attempts to recover or constrain the deeper substrate route, records negative and partial results, and preserves the history of the broader research program. Yet its role in public presentation is now different from what it would have been at an earlier stage.

The bridge archive is no longer the condition for saying that the repository has a theory statement. The theory statement is already given by Æther-flow Interpretation of Relativity as the exact-closure package. The archive is therefore downstream and optional. It should be read as foundational continuation answerable to the exact-closure benchmark, not as the front door to the project.

Remark 3. *If a future bridge route succeeds, it must recover the same exact relativistic benchmark already fixed by the flagship package. If a future bridge route fails, the project contracts back to the exact-closure theory rather than back to ontology alone. At the current stage, the bounded primitive-reservoir line has already ended in a scoped no-go on its frozen bridge object, so any resumed derivational continuation must begin on a genuinely new line with an explicit observer-localizing law or equivalent new structural principle. At minimum, that new law must say how substrate structure maps to observer spacetime fields, how the operative metric is generated locally, and how the Einsteinian observer equation together with relativistic recovery arise from that same local structure.*

9. Research Provenance and Collaboration Method

The repository should also be read honestly as AI-assisted theoretical research under sustained human direction. The original conceptual direction came from the user. GPT-5.4 assisted with drafting, mathematical exploration, branch screening, organization, and revision. The resulting package is therefore neither a purely conventional solo draft nor an autonomous AI production.

This provenance matters for transparency, but it does not change the intellectual standard to which the work should be held. The manuscripts should be judged by their assumptions, equations, internal logic, and honesty about what remains open. AI assistance does not convert speculative steps into established physics, and it does not remove the need for expert human review. Its role here has been methodological support within an explicitly documented collaboration.

10. How the Package Should Be Reviewed

At public-release stage, the most useful technical review focuses on five questions.

1. Is the claim boundary stable and explicit throughout the package?
2. Do the equations and prose say exactly what is established, and no more?
3. Are the package roles of overview, anchor, and companion manuscripts clear and non-overlapping?
4. Does the presentation distinguish properly between the flagship exact-closure package and the downstream bridge archive?
5. Does the presentation make equally clear that the current bounded primitive-reservoir result is a scoped no-go on a frozen bridge object, so any resumed derivational work would require a genuinely new observer-localizing law rather than another same-output relay?

Readers who need the shortest technically disciplined path should begin with the overview and then the exact-closure note. Readers who want the modular full statement should continue through foundations, dynamics, consistency, relativistic recovery, and flow geometry in that order. Only after that should the bridge archive be used for broader foundational context.

11. Conclusion

The repository now has a public-facing flagship theory package. The positive result is that Æther-flow Interpretation of Relativity can already be presented as a disciplined exact relativistic theory statement built on the Æther / Æther-flow ontology, with the effective gravitational dynamics adopted exactly as those of GR. The scope of that result is the overview-first exact-closure package and its companion manuscripts, not a completed first-principles substrate derivation.

The next honest burden is therefore twofold. First, public release materials should continue to package, review, and present this exact-closure package as the flagship result already achieved, with the present article serving as a release-facing synthesis rather than a reordering of the scientific core, and with the overview-first order kept explicit and easy to audit. Second, any future foundational continuation should remain clearly subordinate to that package, should state without ambiguity that deeper derivation from substrate variables remains open work rather than an achieved public claim, and should begin only if a genuinely new observer-localizing law or equivalent new structural principle is ready to be stated.